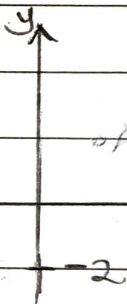


(a)



Let A be the initial point, and B be the terminal point.

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$(-1, 2) = (-1, -2) - \vec{OA}$$

$$\vec{OA} = (-1, -2) - (-1, 2) = (0, -4)$$

$\therefore$ Initial point of vector $-i + 2j$ is $(0, -4)$

$$(b) \quad x(1, -3) + y(-1, 4) = (1, 0)$$

$$x - y = 1 \dots (i)$$

$$-3x + 4y = 0 \dots (ii)$$

From equation (i), $x = 1 + y$

Substituting x for $(1+y)$ in equation (ii):

$$-3(1+y) + 4y = 0$$

$$-3 - 3y + 4y = 0$$

$$-1y = -1 \Rightarrow y = 1$$

Substituting y for 1 in equation (i):

$$x - 1 = 1 \Rightarrow x = 2$$

$$\therefore 2(1, -3) + 1(-1, 4) = (1, 0) \quad \therefore 4(1, -3) + 3(-1, 4) = (1, 0)$$

03

Write on both sides of the paper

Question.....2.....

(8) $\vec{v} \cdot \vec{w} = |\vec{v}| |\vec{w}| \cos \theta$
 $(1, -1, 2) \cdot (2, 1, -1) = (\sqrt{1^2 + (-1)^2 + 2^2}) (\sqrt{2^2 + 1^2 + (-1)^2}) \cos \theta$
 $1 \times 2 + -1 \times 1 + 2 \times -1 = 6 \times 6 \cos \theta$
 $-1 = 36 \cos \theta$
 $\cos \theta = \frac{-1}{36} = -0.0278$
 $\theta = \cos^{-1}(-0.0278) = 91.59^\circ$

Projection of $\vec{v}$ on $\vec{w} = \frac{\vec{w} \cdot \vec{v}}{|\vec{w}|}$
 $\frac{2 - 1 - 2}{6} = -0.1667$

(a) $\underline{v} \cdot \underline{w} = |\underline{v}| |\underline{w}| \cos \theta$
~~(1, -1, 2) \cdot (2, 1, -1) = (\sqrt{1^2 + (-1)^2 + 2^2}) (\sqrt{2^2 + 1^2 + (-1)^2}) \cos \theta~~
 $-1 = \sqrt{6} \sqrt{6} \cos \theta$
 $-1 = 6 \cos \theta$
 $\cos \theta = -\frac{1}{6} = -0.1667$

~~11~~ Projection of $\underline{x}$ on $\underline{w} = \frac{\underline{w} \cdot \underline{x}}{|\underline{w}|}$. $\underline{w} \cdot \underline{x} = -1$
 $|\underline{w}| = \sqrt{6}$
 $= \frac{-1}{\sqrt{6}} = -\frac{\sqrt{6}}{6}$

(b) Vector of the line is $(4, 2, -1) - (-1, 3, 7) = (5, -1, -8)$
 Vector equation of the line is $(x, y, z) = (-1, 3, 7) + \lambda(5, -1, -8)$

$$\begin{aligned} x &= -1 + 5\lambda \\ y &= 3 - \lambda \\ z &= 7 - 8\lambda \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Parametric equations}$$

 Symmetric equations: $\frac{x+1}{5} = \frac{y-3}{-1} = \frac{z-7}{-8}$

Write on both sides of the paper

Question... 3

(a) Let (x, y, z) be a ^{general} point on the plane.

A general vector on the plane will be $(x, y, z) - (3, -7, 2) = (x-3, y+7, z-2)$

$$(x-3, y+7, z-2) \cdot (2, 1, -3) = 0$$

$$2(x-3) + 1(y+7) + -3(z-2) = 0$$

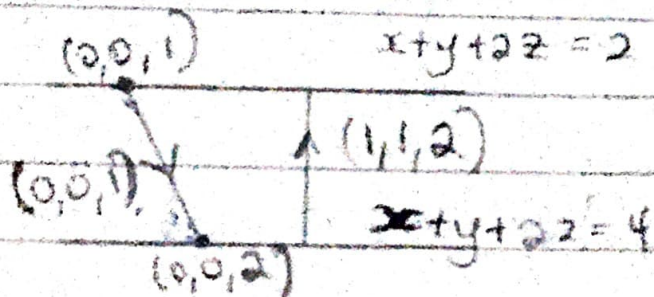
$$2x - 6 + y + 7 - 3z + 6 = 0$$

$$2x + y - 3z - 6 + 7 + 6 = 0$$

$$2x + y - 3z + 7 = 0$$

03

(b)



$$(0, 0, 2) - (0, 0, 1) = (0, 0, 1)$$

Distance between the two planes

$$= \frac{(1, 1, 2) \cdot (0, 0, 1)}{|(1, 1, 2)|} = \frac{2}{\sqrt{6}} = 0.816$$

03

Write on both sides of the paper

Question... 4

$$(i) s|_{t=2} = 40 \times 2 - 10 \times 2^2 = 80 - 40 = 40 \text{ m}$$

$$(ii) v = \frac{ds}{dt} = 40 - 20t$$

$$v|_{t=1} = \left. \frac{ds}{dt} \right|_{t=1} = 40 - 20 \times 1 = 40 - 20 = 20 \text{ m/s}$$

$$(iii) s = 40t - 10t^2$$

$$30 = 40t - 10t^2 \quad \begin{matrix} 3 & -3 & -1 \\ & -4 & \end{matrix}$$

$$10t^2 - 40t + 30 = 0$$

$$t^2 - 4t + 3 = 0$$

$$t^2 - t - 3t + 3 = 0$$

$$t(t-1) - 3(t-1) = 0$$

$$(t-3)(t-1) = 0$$

$$t=3 \text{ or } t=1$$

∴ The instant at which the stone is 30m high is when $t=3$ secs and $t=1$ seconds.